United International University
Department of Computer Science and Engineering

CSE 3313: Computer Architecture
Midterm Examination: Spring 2026
Total Marks: 30 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all questions. Numbers to the right of the questions denote their marks.

1. Suppose a processor P has three instruction classes A, B, and C with CPI values 2, 1, and 3.respectively. The number of instructions(IC) in P is as follows:

Programs	A	B	C
P(IC)	100	110	116
Q(IC)	80	120	195

Consider that both programs run on the same device with a clock frequency of 2 GHz.

(a) Identify which instruction class contributes the most delay in executing the program. [2]

(b) If execution time of class C is improved by 3x, find the overall speedup of program P. [4]

(c) If program Q must achieve the average CPI of P1 then what should be the number of instructions (IC) for instruction class C? [4]

2. (a) Consider the following C function. The arguments will be found at \$a0-\$a1. Convert the code to the corresponding MIPS assembly instructions by assigning registers according to your requirement. [7]

```
int main()
{
    int a = 25, b = a % 16;
    int c = ~(a | 5);
    int d = ~ b;
    if ( c >= d ) { int e = myfunction(c,d); }
    else { int e = myfunction(a,b); }
}
int myfunction ( int y, int x )
{
    return x-y;
}
```

Code for 2 (a)

(b) Observe the following MIPS code and convert it into machine code. Conversion of the machine code to binary is not required. [3]

```
100: LOOP
104: slt $t0, $s1, $s2
108: beq $t0, $zero, END
112: lw $t1, 32 ($s0)
116: sll $t2, $t1, 10
120: j LOOP
124: END
128: add $v0, $a0, $ra
```

Code for 2 (b)

(c) Assume that processor P has 1000 registers. The size of a MIPS instruction is 32 bits, and 6 bits are reserved for the opcode. Find out the second largest memory location that the processor can operate with.

For this processor P, can we use this instruction `andi $s0, $s1, 1023`? Explain. [4+2]

3. Multiply two numbers A by B using the multiplication algorithm, where A = 9 and B = 8. [4]

Instruction	Opcode	Function Code
add	0	32
sub	0	34
lw	35	-
sw	43	-
and	0	36
or	0	37
nor	0	39
andi	12	-
ori	13	-
sll	0	0
srl	0	2
beq	4	-
bne	5	-
slt	0	42
j	2	-
jr	0	8
jal	3	-
addi	8	-

Table 1: MIPS Machine Codes

Name	Register Number
$zero	0
$at	1
$v0-$v1	2-3
$a0-$a3	4-7
$t0-$t7	8-15
$s0-$s7	16-23
$t8-$t9	24-25
$k0-$k1	26-27
$gp	28
$sp	29
$fp	30
$ra	31

Table 2: MIPS Registers